

RAHUL RAVI VK

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SUMMARY

Instrumentation & Controls Engineer with **over 6 years** of experience deploying and commissioning control systems in safety-critical process plants across DCS and PLC platforms. Widening my expertise into robotics, machine vision, and Industrial AI through M.Eng. in Robotics at the University of Maryland. Seeking to design and deploy advanced automation and robotics solutions.

WORK EXPERIENCE

Research Intern (Robotics & Machine Vision) | *Telehealth and Robotics Lab, Children's National Medical Center, Washington, D.C.*
May 2026 – Aug 2026

- Integrate robotic systems and smart sensing under Dr. Reza Monfaredi (Sheikh Zayed Institute): a smart wheelchair for children with Cerebral Palsy and UR5 tool-sorting robot for part identification.

Assistant Manager, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* May 2023 – Jul 2025

- Cut equipment downtime **50%** and false trips **90%** by leading automation and predictive-maintenance programs across **Honeywell Experion PKS C300 DCS** and **Siemens PLC** platforms; engineered pump speed-control logic and PLC bistable-relay emulation.
- Owned automation projects concept-through-commissioning: controls design, PLC sequencing, loop checking, FAT/SAT, and operator handover, coordinating cross-disciplinary Mechanical, Electrical, Production, and Civil teams through annual turnarounds with zero missed restart deadlines.

Senior Engineer, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* Oct 2019 – Apr 2023

- Sustained **99.5%+** availability of **Honeywell Safety Manager (ESD)** and Experion DCS across **4000+ control loops** via RAID5 storage administration and one-to-one spare-station redundancy.
- Reduced process downtime **30%** through hands-on commissioning, PID retuning, and actuator nonlinearity analysis, including a 5-hour online tuning campaign that stabilized a safety-critical loop with zero production interruption.

Graduate Engineer Trainee, Instrumentation & Controls | *KRIBHCO Fertilizers Ltd, Gujarat, India* Oct 2018 – Oct 2019

- Installed, calibrated, and validated 20+ transmitter and sensor types to $\pm 0.075\%$ FS across 4–20 mA / HART loops; commissioned field wiring, PLC sequencing, and safety interlocks on live process lines.

PROJECTS

Air Dryer Control Panel Redesign & Commissioning *Siemens PLC, TIA Portal, Ladder Logic*

- Redesigned and built an in-house PLC control panel replacing worn bistable-relay and mechanical-timer components, eliminating mechanical failure points from the air dryer regeneration path while preserving proven operator-familiar logic.
- Executed a single-day cutover within the 8-hour maintenance window across Mechanical, Electrical, Production, and Civil teams—transitioning to modern PLC-based control with zero production loss.

TurtleBot4 Real-Time Perception Pipeline *ROS 2, OpenCV, Vision, Smart Sensing*

- Engineered a ROS 2 vision pipeline for real-time part/object detection and decision-making: logo/object identification (SIFT + RANSAC), dynamic obstacle detection, and path tracking, all running on-robot under latency constraints.
- Benchmarked deep-learning against classical vision and selected the approach that met real-time deployment constraints, the same trade-off analysis production inspection systems demand.

Autonomous Block Retrieval Robot *OpenCV, Raspberry Pi, Arduino, PID*

- Architected a fully autonomous robot from bare hardware to detect, retrieve, and deliver objects in sequence, **9/9 items delivered in 7:21** (perfect score); integrated vision-based detection, PID visual servoing, gripper actuation, and sensor fusion for closed-loop pick-and-place.

Pump Torque-Converter Speed Control Modernization *Honeywell DCS, Cascade PID, Pepperl+Fuchs*

- Consolidated a legacy dual-cascade pump controller into the DCS and replaced an obsolete speed sensor with a modern Pepperl+Fuchs unit; performed online PID retuning of both level and speed loops during annual shutdown with no permanent downtime.

EDUCATION

M.Eng. in Robotics | University of Maryland, College Park Sep 2025 – May 2027

GPA: 3.9/4.0 | **Coursework:** Perception, Planning, Control of Robotic Systems, Autonomous Robots, Robot Modeling, C++ programming, Industrial AI, ML for Perception & Sensing, Stochastic Control

B.Eng. in Instrumentation & Control | L.D. College of Engineering, Gujarat, India Aug 2014 – May 2018

GPA: 3.3/4.0 | GATE All India Rank 96

TECHNICAL SKILLS

- Automation & Controls:** Honeywell Experion PKS C300 DCS, Siemens PLC, SCADA, Safety Systems (ESD), PID/cascade control, FAT/SAT, commissioning, loop checking
- Robotics & Integration:** ROS 2, MoveIt 2, Gazebo, Nav2, URDF/Xacro, motion planning (A*, RRT*), sim-to-real deployment
- Vision & AI/ML:** Machine vision, smart sensing, object/part detection, deep learning, sensor fusion, real-time inference (OpenCV, PyTorch)
- Embedded & Hardware:** Raspberry Pi, Arduino, NVIDIA Jetson, IMU/encoder/ultrasonic sensors, UART/I2C, 4–20 mA / HART
- Programming & Tools:** Python, C++, MATLAB/Simulink, Git, Docker, SolidWorks

CERTIFICATIONS

Siemens – Basics of PLC & SCADA | Honeywell – Experion PKS C300 Fundamentals